

Agent READMEs: An Empirical Study of Context Files for Agentic Coding

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Agentic coding tools receive goals written in natural language, break them down into specific tasks, and write or execute code with minimal human intervention. Central to this process are agent context files (e.g., AGENTS.md and CLAUDE.md) that provide persistent, project-level instructions. In this paper, we conduct the first large-scale empirical study of 2,303 agent context files from 1,925 repositories to characterize their structure, maintenance, and content. We find that these files are not static documentation but complex, difficult-to-read artifacts that evolve like configuration code through frequent, small additions. Our content analysis of 16 instruction types shows that developers prioritize functional context, such as test procedures (75.9%), implementation details (70.8%), and architecture (68.1%). We also identify a significant gap: non-functional requirements such as security (14.8%) and performance (14.5%) are rarely specified. These findings indicate that while developers use context files to make agents functional, they provide few guardrails to ensure that agent-written code is secure or performant, highlighting the need for improved tools and practices.

CCS Concepts: • **Software and its engineering** → **Automatic programming; Software configuration management and version control systems; Software evolution; Maintaining software.**

Additional Key Words and Phrases: Agentic Coding, Autonomous Programming, Documents

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1 INTRODUCTION

“The hottest new programming language is English.”
 — Andrej Karpathy (Founding member of OpenAI)

Large Language Models (LLMs) are transforming software development from editing code by hand to instructing Artificial Intelligence (AI) agents in natural language [28, 35]. This novel approach, termed *Agentic Coding*, interprets natural language goals, decomposes them into subtasks, and autonomously plans and writes code with minimal human intervention. However, the autonomy of these agents hinges on a critical, often overlooked factor: *context*. For an agent to work for a software project, adhere to architectural patterns, and follow team conventions, it requires not only access to source code but also explicit guidance. The effectiveness and safety of this workflow depend not only on model quality but also on how projects communicate their architecture, constraints, and conventions to these agents.

To address this, a new class of software artifact has emerged: *agent context files*. These are specialized files (e.g., `AGENTS.md`) that work like a README for agents¹ and define how the agent should behave within a given software project. These documents specify project-specific knowledge, architectural overviews, build and test commands, coding conventions, and explicit rules for interacting with tools and external services. They define the agent’s role, describe the system architecture, record build and test commands, and encode coding standards and operational rules. Modern coding tools such as Claude Code, OpenAI Codex, and GitHub Copilot load these files at the start of a session and use them as persistent context when planning and executing changes. In practice, the quality of these agent context files largely determines how well an agent can understand a repository, follow team practices, and perform non-trivial maintenance tasks.

Prior work on LLM-assisted software engineering has focused on model capabilities, interaction patterns, and immediate prompts, for example, code generation [26, 31], and the impact of natural language inputs on generated code quality [41], automatic program repair [11, 78, 85], and multi-agent collaboration [83]. Researchers have also begun to study context engineering [26, 63] and prompt management in software repositories [34, 64]. However, the concrete artifacts that teams already use in the wild to steer coding agents (namely agent context files) remain underexplored. Official tool documentation gives only high-level guidance (e.g., “describe architecture and workflows”), leaving developers to design these agent context files by trial and error. We currently lack basic empirical evidence about what these files look like, how they evolve over time, and what instructions they actually contain.

To bridge this gap, we conduct a large-scale empirical study of agent context files. Building on our prior work [14] which studied 253 context files for Claude Code, we extend the scope to 2,303 context files from 1,925 repositories across three agentic coding tools: Claude Code, OpenAI Codex, and GitHub Copilot. We aim to uncover the structural patterns, maintenance habits, and instructional strategies that developers currently employ. Our study focuses on these research questions (RQs):

RQ₁: What are the characteristics of agent context files? Understanding readability and structure is essential for creating maintainable agent configurations. Agent context files are generally long and difficult to read, yet follow a consistent shallow hierarchy with a single top-level heading and most content organized under H2 and H3 sections.

¹<https://agents.md>

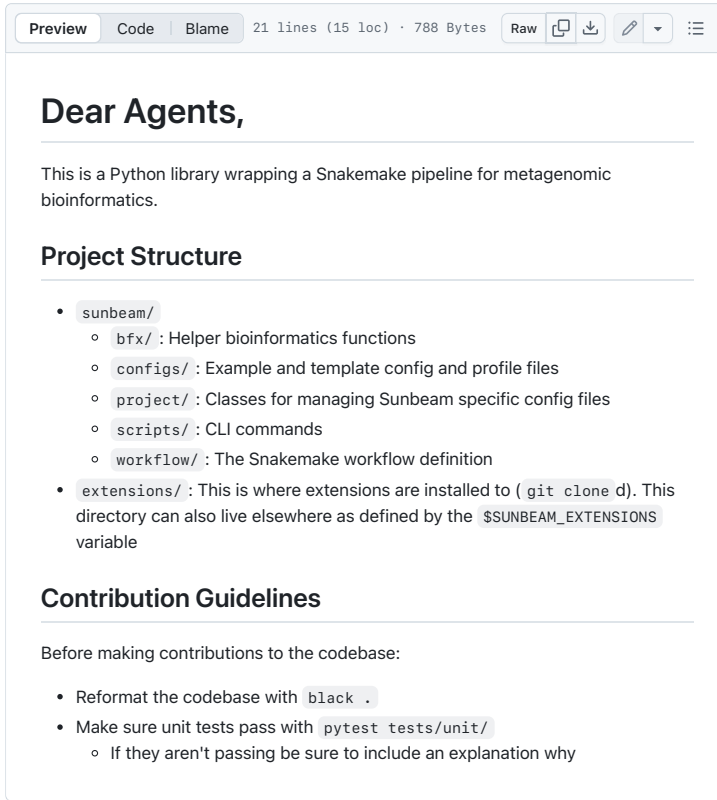


Fig. 1. Example of agent context files

RQ₂: How often do developers maintain agent context files? It is unclear whether agent context files behave like static documentation or like evolving configuration in the codebase. Agent context files are actively maintained in short bursts, evolving through incremental additions rather than deletions, behaving as living configuration artifacts rather than static documents.

RQ₃: What instructions are included in agent context files? Identifying what context developers prioritize reveals the current focus and blind spots of agentic coding in real-world projects. Instructions are heavily skewed toward functional operations (e.g., Build and Run, Implementation Details), while critical non-functional requirements like Security and Performance are rare.

RQ₄: To what extent can instructions in agent context files be classified automatically? Manual analysis of agent context files' content is unscalable, making it critical to understand if this classification can be automated for future monitoring. Automatic classification is highly effective (0.79 F1-score) for concrete functional topics (e.g., Testing and Architecture), but it struggles with abstract or nuanced topics (e.g., Maintenance).

Replication Package. To facilitate replication and further studies, we provide the data used in our replication package.²

²<https://github.com/woraamy/Agent-Context-File-Analysis>

Paper Structure. The remainder of this paper is organized as follows. Section 2 presents the motivating examples. Section 3 details our study design, data collection, and analysis methodology. Section 4 reports the empirical findings. Section 5 summarizes the key insights and discusses the implications of our study. Section 6 reviews related work. Section 7 outlines threats to validity, and Section 8 concludes the paper.

2 MOTIVATING EXAMPLES

Agentic coding tools can be configured through specialized Markdown files that define how AI coding assistants should operate within specific projects. Common examples include `CLAUDE.md` for Claude Code or `AGENTS.md` for OpenAI Codex. We refer to these artifacts as *Agent Context Files*. Agent context files serve as a persistent long-term memory for the agent. By documenting project-specific context such as architectural patterns, testing commands, and coding conventions, these files spare developers from repeatedly explaining the same rules in each session. When stored in version control, they help the AI maintain a consistent understanding of the project's requirements, even as the codebase evolves.

The structure and content of agent context files vary significantly across projects, reflecting diverse development needs and practices. Consider two real-world examples that illustrate this diversity. The Python bioinformatics library (shown in Figure 1) contains a minimal `AGENTS.md` file of just 21 lines with 3 sections, providing basic instructions about running commands, code style, and directory structure. In contrast, an agent-tool development project could feature a comprehensive 329-line file with 74 distinct sections, covering project overview, architecture, development workflow, security and performance considerations, testing strategy, etc.³ These examples demonstrate that agent context files are not standardized or template documents but rather adaptable configuration mechanisms that developers customize based on project complexity, team size, and domain-specific requirements.

Despite their importance, official documentation for agent context files remains limited. For example, the official documentation for Claude Code indicates only that agent context files can be used to share instructions for the project, such as project architecture, coding standards, and common workflows. This fragmentation and lack of explicit, standardized guidance can delay developers' ability to effectively define and leverage the agent's behavior, creating a substantial barrier to exploiting the full potential of agentic coding and potentially leading to inconsistencies in agent performance and a steeper learning curve for integration. Consequently, this gap in practical guidance for the creation of agent context files served as a primary motivation for our current research. We aim to address this challenge by systematically investigating existing agent context files within open-source repositories. Our study is specifically designed to infer common structural patterns, typical instructions, and general practices in how developers configure and maintain these crucial agent context files.

3 METHODOLOGY

To expand upon our previous work [14], this study systematically collects and analyzes agent context files from open-source repositories for three agentic coding tools: Claude Code, OpenAI Codex, and GitHub Copilot.

Figure 2 illustrates the data collection pipeline. We identify repositories that use agentic coding tools through the AIDev dataset [35], which provides a curated list of repositories where agentic coding tools contribute to development. From this dataset, we select 8,370 repositories with at

³<https://github.com/antimetal/system-agent/blob/3552a87720e0e16a9c05681b1e549c9a73921ade/CLAUDE.md>

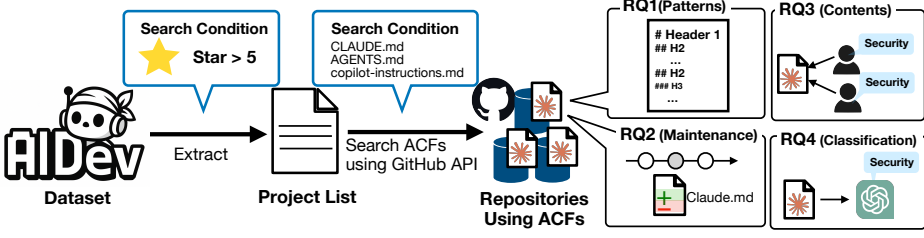


Fig. 2. Overview of our data collection pipeline.

Table 1. Overview of the repositories hosting agent context files (ACFs). We search for the specific filename recommended by the official documentation of each agent.

Agent	File Name	# Files	Median # Stars	Median # Forks	Avg. Days Until ACFs Adoption
Claude Code	CLAUDE.md	922	51.0	10.5	994.9
OpenAI Codex	AGENTS.md	694	56.0	12.0	1100.3
GitHub Copilot	copilot-instructions.md	687	96.0	31.0	1567.9

least 5 GitHub stars to exclude toy projects. We then use the GitHub API⁴ to scan the root directory of each selected repository for files whose filenames follow the official naming convention specified in each agent’s documentation. Specifically, we search for CLAUDE.md, AGENTS.md, and copilot-instructions.md using case-insensitive filename matching. This process yields 922 Claude Code files, 694 OpenAI Codex files, and 687 GitHub Copilot files across 1,925 repositories. Table 1 summarizes the statistics of their hosting repositories, including their median stars, median forks, and the average number of days from repository creation until the first adoption of agent context files. Using this dataset, we address four research questions.

4 RESULTS

In this section, we provide the motivation, approach, and findings for each of our research questions (RQs).

4.1 RQ₁: What are the characteristics of agent context files?

4.1.1 Motivation. Agent context files serve as the initial instructions an agent reads before execution. Understanding their characteristics (such as size, readability, and organizational structure) in real-world projects is crucial for helping developers write effective agent context files. These attributes, much like in traditional software artifacts, directly impact usability, maintainability, and clarity. For instance, agent context file size and the amount of context provided can correlate with task performance and computational cost [36]. Readability is also critical: instructions that are hard to read can be misunderstood by agents or incorrectly modified by developers, leading to unintended behavior [50]. For example, higher readability has been associated with better outcomes in issue resolution tasks [15].

Prior work on traditional software documentation shows that developer-focused documents typically employ hierarchical section structures [72]. Empirical studies of project documentation (e.g., README files) also find that early versions tend to be minimal, focusing on basic usage [22].

⁴<https://docs.github.com/en/rest>

Table 2. FRE score interpretation.

FRE Score	School Level	Reading Difficulty	Example Text Type
< 0	Highly Specialized	Extremely Difficult	Academic or legal documents
[0, 30)	College Graduate	Very Difficult	Technical reports, some legal documents
[30, 50)	College	Difficult	High school-level material
[50, 60)	10th to 12th grade	Fairly Difficult	Consumer information, most web content
[60, 70)	8th and 9th grade	Plain English	Average newspapers
[70, 80)	7th grade	Fairly Easy	Magazines
[80, 90)	6th grade	Easy	Popular magazines
[90, 100)	5th grade	Very Easy	Children’s books
≥ 100	Pre-school	Extremely Easy	Picture books, basic primers

However, agent context files such as CLAUDE.md are a new class of documentation designed specifically for coding agents. To the best of our knowledge, no prior work has systematically analyzed the structural characteristics of agent context files (e.g., how many levels of instructions they contain). Furthermore, the structural elements of these files, such as headers, likely serve a dual purpose. Syntactically, headers act as semantic anchors that help agents locate relevant sections. From a human perspective, consistent sectioning reflects standard documentation conventions that make agent context files easier for developers to maintain. Therefore, we investigate not only their size and readability but also their organizational patterns to understand how this dual audience is currently accommodated in practice.

4.1.2 Approach. We analyze three aspects of agent context files: size, readability, and structural organization. Agent context files size is measured by counting words using the regular expression “\w+”. This pattern matches maximal sequences of word characters (letters, digits, and underscores), and each match is treated as a single word token. Readability is measured with the Flesch Reading Ease (FRE) metric, following prior work on LLM-related texts [13, 15]. We calculate FRE metric programmatically using the Textstat⁵ Python library. The metric operates on two fundamental linguistic assumptions: (1) longer sentences increase cognitive load for the reader, and (2) words with higher syllable counts (polysyllabic words) indicate higher conceptual complexity. The score is calculated using the following formula:

$$\text{FRE} = 206.835 - 1.015 \left(\frac{\text{total words}}{\text{total sentences}} \right) - 84.6 \left(\frac{\text{total syllables}}{\text{total words}} \right)$$

The calculation process follows a rule-based algorithm: the text is first tokenized into sentences and words. Syllable counts are then estimated using standard English syllabification patterns, cross-referenced with the nltk.corpus.cmudict (Carnegie Mellon Pronouncing Dictionary)⁶ for US English to maintain high precision. For words not found in the dictionary, the Pyphen⁷ module is leveraged for algorithmic hyphenation. Table 2 summarizes the interpretation of FRE scores, where higher values indicate easier text.

To capture organizational structure, we extract each Claude Code, GitHub Copilot, and OpenAI Codex file from the cloned repositories and count Markdown headers at each level from H1 (i.e., #)

⁵<https://textstat.org/>

⁶https://www.nltk.org/_modules/nltk/corpus/reader/cmudict.html

⁷<https://github.com/Kozea/Pyphen>

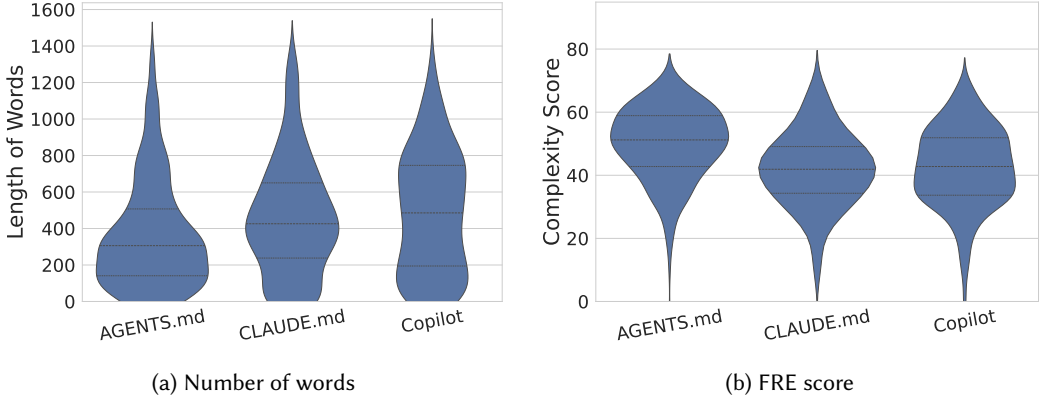


Fig. 3. Distribution of agent context files (a) size and (b) readability.

to H5 (i.e., #####). For example, in Figure 1, “Project Structure” and “Contribution Guidelines” are both H2 headers (indicated by ## in the original Markdown source).

We define headers based on explicit Markdown syntax rather than visual formatting or fonts. Specifically, we use a regular expression to detect lines starting with one or more hash symbols (#) followed by a space. For example, a line starting with ## is classified as an H2 header. Because this syntax is standardized across all context file types, the header levels are inherently comparable across different repositories. This yields the depth and distribution of sections within these agent context files.

Furthermore, we analyze the complexity and scale of the programming tasks solved by the AI agents after adopting a context file. We restrict our analysis to programmatic source code and leverage the Lizard⁸ library to calculate two metrics: Cyclomatic Complexity and Net Lines of Code (*nloc*). For Claude Code and GitHub Copilot, we extract commit-level data locally using Pydriller.⁹ For OpenAI Codex, we use the GitHub REST API¹⁰ to isolate AI-generated commits based on pull request metadata and branch naming conventions. To ensure clear attribution, we exclude commits in which multiple AI tools were used simultaneously, focusing only on instances where a single agent can be definitively identified.

To identify statistically significant differences across coding agents, we perform the Mann-Whitney U test [39] at a significance level of $\alpha = 0.05$. We compute Cliff’s delta d [37] effect size to quantify the difference based on the following thresholds [57]:

$$\text{Effect size} = \begin{cases} \text{negligible,} & \text{if } |d| \leq 0.147 \\ \text{small,} & \text{if } 0.147 < |d| \leq 0.33 \\ \text{medium,} & \text{if } 0.33 < |d| \leq 0.474 \\ \text{large,} & \text{if } 0.474 < |d| \leq 1 \end{cases} \quad (1)$$

4.1.3 Findings. Agent context files for Claude Code and GitHub Copilot are substantially longer than those for OpenAI Codex and this gap is independent of task complexity. Figure 3a shows that GitHub Copilot (median 535.0 words) and Claude Code (median 485.0 words) use similarly long agent context files, with no statistically significant difference between them. Both

⁸<https://github.com/terryyin/lizard>

⁹<https://pydriller.readthedocs.io/en/latest/>

¹⁰<https://docs.github.com/en/rest>

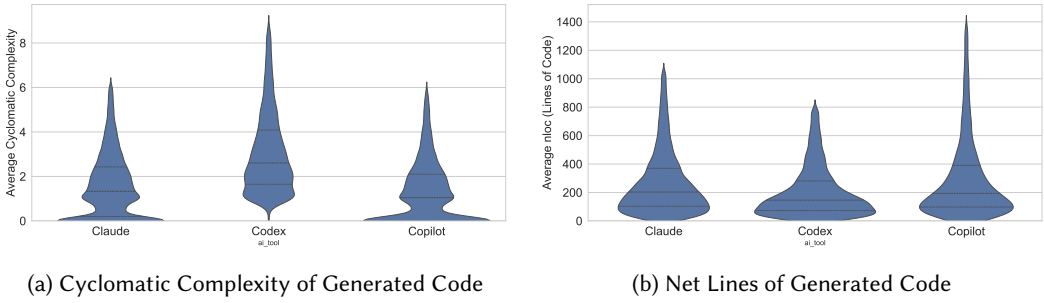


Fig. 4. Distribution of agent context files task difficulty (a) complexity and (b) volume.

are significantly longer than OpenAI Codex agent context files (median 335.5 words), although the effect sizes are small (Cliff's delta $d = 0.21$ and $d = 0.22$, respectively). This difference in context-file length cannot be explained by task difficulty. As shown in Figure 4a, repositories using OpenAI Codex have the highest cyclomatic complexity (median 2.00), statistically higher than repositories using Claude Code (median 1.37), although the effect size is negligible ($d = -0.14$) and GitHub Copilot (median 1.00, $d = -0.26$). However, Figure 4b shows that these tasks produce fewer net lines of code than those generated with GitHub Copilot and Claude Code. The differences in code volume are statistically significant, with GitHub Copilot and Claude Code generating larger code volumes than OpenAI Codex ($d = 0.27$ and $d = 0.37$, respectively). These findings suggest that agent-specific documentation conventions, potentially shaped by prompting styles, default context management, and recommended usage practices, are the primary drivers of these differences.

Agent context files are generally difficult to read, with Claude Code the hardest and OpenAI Codex the easiest. Figure 3b shows that Claude Code agent context files are among the most difficult to read, with a median complexity score of 42.39. This places them in the “difficult” category, typical of high school-level material (see Table 2). GitHub Copilot files, which are “difficult” (median score 44.39), are statistically similar to Claude Code's with a negligible effect size ($d = 0.06$). In contrast, OpenAI Codex files are the easiest to read of the group, with a median complexity score of 51.41. This places them in the “fairly difficult” category, which is typical of 10th to 12th-grade consumer information. Based on the statistical analysis, OpenAI Codex files are significantly easier to read than both Claude Code's (medium effect size, $d = 0.4$) and GitHub Copilot's (small effect size, $d = 0.33$).

To manually validate the FRE scores, one inspector reviews a sample of 100 agent context files and assigns each file a perceived readability label (Easy, Normal, or Hard) without seeing its computed score. The Spearman rank correlation between the labels and the FRE scores is $\rho = 0.19$ ($p = 0.05$), indicating a weak positive relationship. Although the FRE scores and human judgments trend in the same direction, they do not align closely. Given this low correlation, we conduct a per-group inspection to examine what the FRE bands capture. We sample 10 files from each band defined in Table 2: Easy (FRE ≥ 80), Normal ($50 \leq \text{FRE} < 80$), and Hard (FRE < 50), and inspect their content to identify the writing patterns associated with each band.

Files in the Easy group consist primarily of short, single-clause instructions organized as flat bullet lists with little connective prose. Tool names and command strings often appear as inline code tokens, which FRE treats as short, one-syllable units and therefore inflates the score. Files in the Hard group show the opposite pattern: dense prose describing system architecture, technology stacks, and component relationships. Sentences are longer and often contain acronyms, compound noun phrases, and repository-specific technical terms. Files in the Normal group exhibit the most common

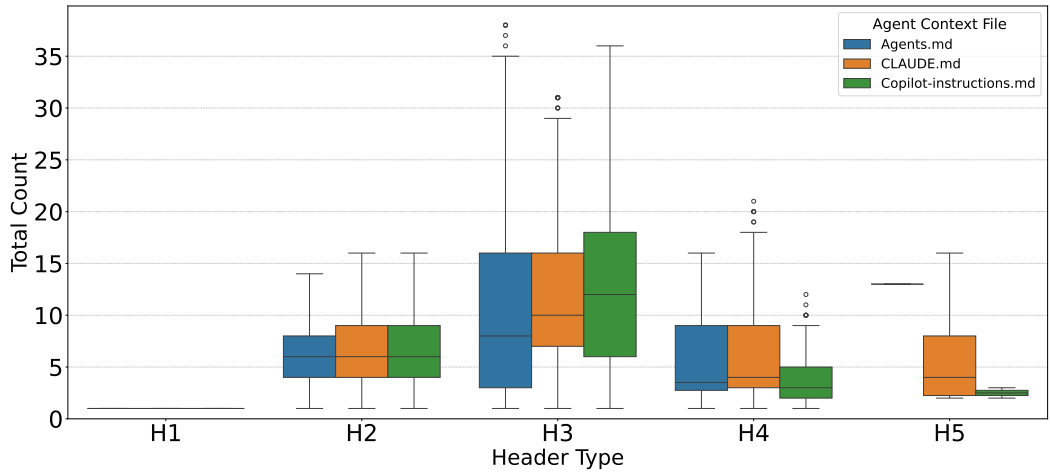


Fig. 5. Distribution of agent context files' Markdown Headers

structure in the corpus, combining hierarchical section headings with short operational list items and brief contextual paragraphs. The prose introduces the project and its key components, while the list items provide concise instructions. This inspection reveals a counterintuitive relationship between FRE scores and informational richness. High-scoring (Easy) files are not necessarily the most useful because their scores largely reflect list-heavy writing rather than actual clarity. In contrast, low-scoring (Hard) files often provide the most complete architectural context, which agents need to understand a project's structure and constraints. This finding suggests that low FRE scores in our dataset partly reflect technical vocabulary and document structure rather than true difficulty for experienced developers.

Agent context files follow a consistent, shallow hierarchy centered on a single top-level heading with most structure at H2 and H3. Figure 5 shows that the structure is almost always a single, top-level H1 heading, with a median count of 1.0 for all agent context files (with no statistically significant difference), suggesting developers treat each file as a unified document. From this primary heading, the agent context files branch into a moderate number of H2 headings (median 6 to 7) to define major topics. While all agent context files follow this pattern, GitHub Copilot and Claude Code agent context files provide significantly more granular detail through a greater number of H3 and H4 sub-sections compared to OpenAI Codex. For example, at the H3 level, GitHub Copilot (median 12.0) and Claude Code (median 11.0) are significantly more detailed than OpenAI Codex (median 9.0), with these differences being statistically significant with a small effect size ($d = 0.15$ and $d = 0.17$, respectively).

Deeper levels are uncommon. At the H4 level, Claude Code (median 5.0) is also statistically significantly more granular than GitHub Copilot (median 4.0), though the effect size remains small ($d = 0.17$). Despite these differences in granularity, the preference for a shallow hierarchy is universal. Deeply nested structures (H5 and beyond) are extremely rare. For example, H5 headers appear in only 9 files, and H6 headers are nonexistent, with no significant difference in their infrequent use. This common structural template: a single H1, several H2s, and some H3s or H4s, likely makes the agent context files easier for developers to quickly parse, modify, and maintain.

Answer to RQ₁

Agent context files are generally long and difficult to read, and these characteristics are driven by platform-specific documentation conventions rather than by the difficulty of the underlying programming tasks. Claude Code and GitHub Copilot agent context files are significantly longer than OpenAI Codex, with Claude Code being the hardest to read and OpenAI Codex the easiest. Structurally, they follow a consistent, shallow hierarchy, typically centered on a single H1 heading with content organized primarily under H2 and H3 sub-sections.

4.2 RQ₂: How often do developers maintain agent context files?

4.2.1 Motivation. Prior empirical studies consistently show that traditional software documentation (e.g., README files) is maintained far less actively than source code [20, 22]. Such documentation is often created early in a project's lifecycle and then receives only minor or infrequent updates, often limited to formatting or link corrections [19]. However, prior work has focused on general-purpose documentation written for humans. Agent context files such as `CLAUDE.md` serve a different purpose: they function as operational configurations for AI coding agents and directly influence how these agents behave in the development workflow. It is unclear whether developers treat them like traditional documentation or as live configuration artifacts that evolve alongside the code. This RQ investigates how often developers modify these agent context files and how intensive that maintenance is over time, to determine whether they exhibit the same maintenance inertia as traditional documentation or follow a more active and dynamic evolution.

4.2.2 Approach. To study the maintenance of agent context files, we analyze their commit histories in the cloned repositories from our dataset. For each repository, we identify all commits in which at least one agent context file (`CLAUDE.md`, `AGENTS.md`, or `copilot-instructions.md`) is added, deleted, or modified. For every such commit, we extract (i) the timestamp and (ii) the number of lines added and deleted and (iii) the number of words in added/deleted diff lines. We then aggregate these per-commit measurements to compute the total number of commits that touch each file and the time intervals between consecutive file-related commits. This yields 5,655 commits for Claude Code, 2,767 commits for OpenAI Codex, and 2,237 commits for GitHub Copilot. Finally, we compare the distributions of commit counts, inter-commit intervals, and added/deleted lines across the three coding agents using Mann-Whitney U with Cliff's delta (same as Section 4.1), to assess whether the context files for different agents follow distinct maintenance patterns.

Moreover, to complement the quantitative analysis above and qualify what the observed commits actually contain, we additionally perform a manual qualitative inspection of a random sample of 100 commits drawn from the population of commits that modify an agent context file, explicitly excluding the commit that first introduces the file so the sample reflects post-adoption maintenance. Two inspectors independently label each commit using a multi-label scheme with six categories derived through open coding: `add:instructions/line` and `add:instructions/section` (introducing new directives at line- or section-level granularity, example in Listing 1¹¹), `fix:instructions` (modifying the wording, scope, or constraints of directives that already exist, example in Listing 2¹²), `del:instructions/line` and `del:instructions/section` (removing directives, example in Listing 3¹³), and `maintenance` (non-functional changes such as reformatting or adding architectural notes that do not change agent-facing directives). A single commit can carry multiple labels when

¹¹<https://github.com/TulevaEE/onboarding-service/commit/7ea1e36e0e47cdd7487d16229ce215d8b536488e>

¹²<https://github.com/mastra-ai/mastra/commit/785251b90ff67c5dc3731ea9cf111eb695288121>

¹³<https://github.com/dogmatiq/dogma/commit/854d999e6f376d0a25611b600f9de1f946f346c>

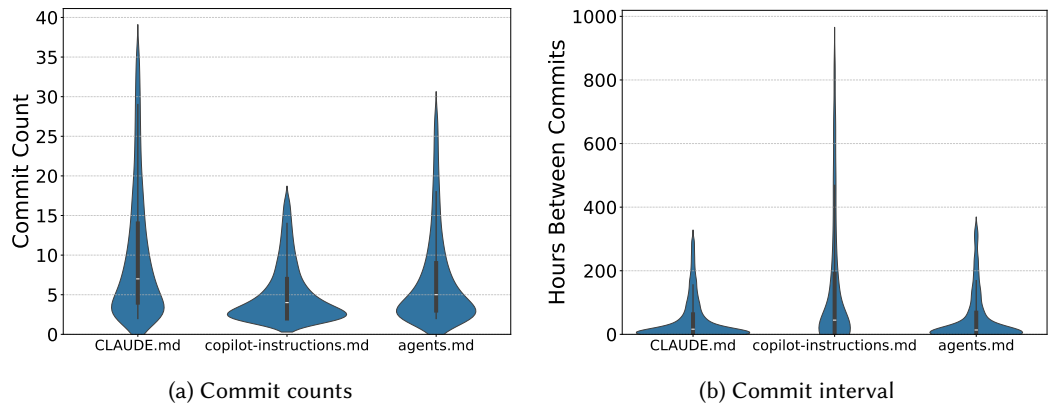


Fig. 6. Distribution of agent context files commit activities.

its diff spans multiple categories. The two inspectors disagreed on 34 of the 100 commits; a third inspector independently resolved all disagreements, and we use the third inspector’s labels as the primary source. The 100 commits yield 139 label assignments, with 33 commits receiving two or more labels. For every labeled commit, we additionally record the time gap (in hours) to the previous context-file commit in the same repository, which allows us to compare when meaningful instruction changes occur.

Listing 1. Example Commit For Adding Instructions

```
1 + #### CRITICAL: Always Write and Run Tests
2 + **Write tests for every change**
3 + **Run tests before committing**
```

Listing 2. Example Commit For Fixing Instructions

```
1 - `pnpm setup` - Install dependencies and build CLI (required first step)
2 + `pnpm run setup` - Install dependencies and build CLI (required first step)
```

Listing 3. Example Commit For Deleting Instructions

```
1 - Reflow paragraphs for readability:
2 - Analyze each paragraph as a whole, not just individual lines.
3 - Fit as many words as you can on each line, up to a hard 80 character limit.
```

4.2.3 Findings. Agent context files are actively maintained through substantive instruction edits, rarely behaving as static documentation or maintenance updates. Figure 6a shows that a majority of Claude Code context files (67.0%) are modified in multiple commits, indicating that they are routinely revisited and refined rather than being written once and forgotten. This pattern also holds, though slightly less frequently, for GitHub Copilot (59.8%) and OpenAI Codex (59.4%). The results of the Mann-Whitney U test show that Claude Code context files receive significantly more commits than both GitHub Copilot (with a medium effect size where $d = 0.35$) and OpenAI Codex (with a negligible effect size where $d = 0.14$). In addition, OpenAI Codex context files are

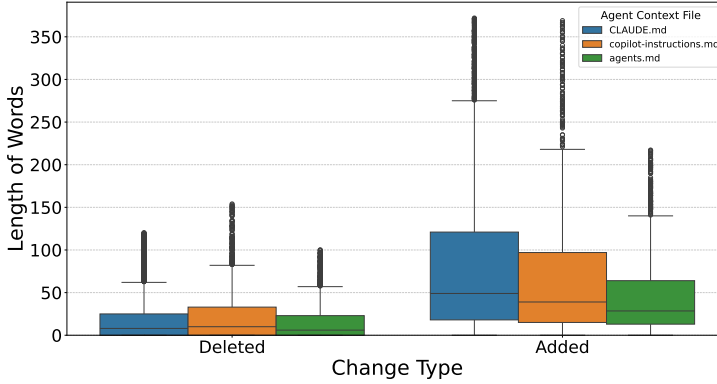


Fig. 7. Distribution of the number of deleted and added words in agent context files commits.

updated more often than GitHub Copilot with a small effect size ($d = 0.18$). Overall, agent context files behave as evolving configuration artifacts and not as static instructions.

The qualitative inspection of 100 sampled post-adoption commits affirms this interpretation. Some form of substantive instruction edit (adding, fixing, or deleting directives) appears in 92% of the sampled commits, while purely non-functional maintenance is the only label in just 8%. Even among files in the short tail of Figure 6a, 42% of the sampled commits occur in files with five or fewer file-specific commits. The work being done is overwhelmingly instruction maintenance rather than decorative, confirming that the active-maintenance pattern reflects real refinement of the agent instructions.

Maintainers update agent context files in short bursts that reflect ongoing development.

Figure 6b shows that the median interval is 68.0 hours (around 3 days) for GitHub Copilot, 23.8 hours (around 1 day) for Claude Code, and 22.3 hours (around 1 day) for OpenAI Codex. These intervals span days rather than minutes or hours, which suggests that updates are not the result of trial-and-error debugging or immediate prompt tuning. If developers were simply trying to fix a prompt that did not work as expected, we would expect a high density of commits within a single session. Instead, these day-scale intervals suggest that developers update context files as they encounter new project requirements or coding conventions over the course of ongoing development. To statistically validate these variations, we applied the Mann-Whitney U test and Cliff's delta for effect size. The results confirm that the update interval distributions differ significantly across all compared pairs ($p < 0.001$). The disparity is most pronounced between GitHub Copilot and OpenAI Codex, which exhibits a small effect size ($d = -0.21$). The comparison between GitHub Copilot and Claude Code also shows a small effect size ($d = 0.18$), while the comparison between OpenAI Codex and Claude Code is negligible ($d = -0.04$).

The manual inspection further strengthens this interpretation by characterizing the timing of each type of commit. If the day-scale intervals were instead driven by prompt debugging, commit types involving revisions to instructions that did not produce the intended agent behavior should be the most tightly clustered category. The sample shows the opposite. Across the 100 labeled commits, the overall median gap to the previous context-file commit is 36.7 hours (42% within 24 hours, 61% within 72 hours), and fix: instructions has the longest median gap of any common label (54.5 hours, only 37.5% within 24 hours and 53.1% within 72 hours). The most temporally clustered categories are instead the deletion labels (del: instructions/line at a median of 16.6 hours and del: instructions/section at 0.8 hours), which are consistent with localized cleanup

rather than iterative prompt tuning. This result is the opposite of what a debugging pattern would produce and reinforces our interpretation of these day-scale intervals as developers correcting and extending instructions as they encounter new requirements during ongoing development.

The evolution of Agent context files is driven by small, incremental additions, while deletions are minimal. Figure 7 shows that deletions are consistently negligible across all agent context file types, with median values less than 15.0 words. In contrast, additions are more substantial and variable. Claude Code updates tend to be the most substantial, with a median of 57.0 words added per commit. This is significantly more words than OpenAI Codex and GitHub Copilot, although the effect size is negligible.

The label distribution in the manual sample confirms this pattern at the commit level: addition labels appear 78 times across 66 of 100 commits (48 line-level, 30 section-level), while instruction modification (`fix: instructions`) appears in 32 of 100 commits, reflecting developers refining existing directives rather than replacing them as a whole. Deletion labels appear in only 19 (15 line-level, 4 section-level). Combined with the results in Figure 7, this suggests that the increase in line counts is not simply due to uneven edit sizes, but instead reflects a meaningful tendency to add new instructions rather than remove existing ones.

Answer to RQ₂

Agent context files are actively maintained and a majority (59% to 67%) are modified in multiple commits, with Claude Code files receiving significantly more commits than others. Maintenance typically occurs in short bursts, with updates often performed about a day apart, and evolution is driven mainly by small incremental additions rather than large-scale deletions or rewrites.

4.3 RQ₃: What instructions are included in agent context files?

4.3.1 Motivation. Prior work shows that clear, structured instructions, such as stepwise task descriptions or templated formats, significantly improve LLM-generated outputs [82], with performance fundamentally shaped by the contextual information provided [43]. Despite this, there is little empirical research on agent context files intended to configure and guide AI agents. RQ3 addresses this by identifying prevalent instruction patterns in these agent context files, revealing how developers structure context to align AI behavior in practice.

4.3.2 Approach. To systematically analyze the instructional content of agent context files, we adopted a qualitative coding approach inspired by grounded theory [10, 60, 66]. Because no established taxonomy exists for instructions in these files, we defined the categories directly based on the data. Following the grounded-theory process, we conducted open coding (initial label generation), axial coding (identifying relationships and consolidating codes), and selective coding (applying and validating the core categories).

Open Coding. We focused on constructing a robust and comprehensive label set. We began by extracting all the H1 and H2 titles from Claude Code files. These headers were used only as seed material for generating candidate labels during open coding. Subsequently, we prompted three popular large language models (LLMs), Claude Opus 4.1, Gemini 2.5 Pro, and GPT-5, to propose candidate labels. One author selected the most appropriate label from these suggestions or created a new label when necessary. The use of LLMs was motivated by findings from prior research [5], and we used LLMs only to generate candidate codes, while all final decisions remained human-curated, consistent with the human-in-the-loop nature of open coding. Two additional authors independently reviewed the resulting label set, yielding 61 initial labels.

Axial Coding. Three inspectors refined these labels into a more general taxonomy using the constant comparison method [60, 66]. Semantically overlapping labels were merged, while finer-grained labels were absorbed into broader categories. For example, Business Domain and Business Logic were merged into System Overview, and Best Practice was incorporated into Implementation Details. This process produced a final taxonomy of 16 categories. Categories with similar surface topics but distinct instructional purposes were intentionally kept separate. The comprehensive label guide can be found in our Appendix A.1.

Selective Coding. We extended our labeling to a broader set of agent context files to validate the explanatory coverage of the 16 core categories and refine their boundaries through systematic application [10, 60, 66]. To ensure a representative sample across platforms, we performed proportional random sampling based on the size of each source dataset: Claude Code (922 repositories, 40.0%), OpenAI Codex (694 repositories, 30.1%), and GitHub Copilot (687 repositories, 29.9%). This resulted in a target sample of 332 files. An automated script filtered non-English files using the langdetect library, replacing them with randomly selected English files from the same dataset until the target sample size of 332 files was reached.

Because a single file often contains multiple types of instructions, we adopted a multi-label coding scheme. Two inspectors independently assigned one or more of the 16 categories to the complete content of each file. To ensure consistency between related categories, we defined explicit decision rules. For example, Documentation and References was reserved for instructions about documentation artifacts (e.g., JSDoc or usage examples), whereas documentation-related coding conventions were assigned to Implementation Details.

To assess the reliability of the taxonomy, two inspectors independently labeled all sampled files. This process produced 2,227 label assignments, with 438 disagreements, corresponding to an agreement rate of 80.3%. A third inspector resolved all disagreements through negotiated consensus [60], resulting in 2,069 final label assignments. The three inspectors had between 4 and 17 years of programming experience.

4.3.3 Findings. The manual classification process identified 16 distinct categories of agent context files. Table 3 presents the distribution of documentation categories, where percentages indicate the proportion of agent context files containing instructions for each category. We identified 16 distinct labels, including two new categories (Maintenance and Debugging) beyond those in our previous work [14]. These additions were necessary to classify cases that did not fit existing categories.

As shown in the table, the most prevalent category was Testing (75.9%), containing procedures related to automated tests. This was followed by Implementation Details (70.8%) with development guidance (e.g., code style) and Architecture (68.1%) describing high-level system design.

While the most prevalent instructions (Testing, Implementation Details, Architecture) address functional aspects, meta-level or non-functional categories like Performance (14.5%), Security (14.8%), and UI/UX (8.7%) appear far less frequently. This pattern suggests that agent context files are primarily optimized to help agents execute and maintain code efficiently rather than address broader quality attributes or user-facing aspects.

Beyond functional categories, we observed notable instances where developers provide contextual information. For example, half of the files contain system overview explanations (i.e., System Overview). Additionally, 24.1% of files (i.e., AI Integration label) explicitly define the agent's role and describe its responsibilities within the project (e.g., reviewers). This indicates that agent context files serve not only as technical guides but also as a means of establishing an AI agent's understanding, responsibilities, and collaborative alignment.

Table 3. Categories, descriptions, and their prevalence in agent context files (ACFs).

Category	Label	Description	% ACFs
General	System Overview	Provides a general overview or describes the key features of the system.	59.0
	AI Integration	Contains specific instructions on the desired behavior and roles of agentic coding, as well as methods for integrating other AI tools.	24.1
	Documentation and References	Lists supplementary documents, links, or references for additional context.	26.8
Implementation	Architecture	Describes the high-level structure, design principles, or key components of the system's architecture.	68.1
	Impl. Details	Provides specific details for implementing code or system components, including coding style guidelines.	70.8
Build	Build and Run	Outlines the process for compiling source code and running the application, often including key commands.	63.0
	Testing	Details the procedures and commands for executing automated tests.	75.9
	Conf. & Env.	Instructions for configuring the system and setting up the development or production environment.	38.9
	DevOps	Covers procedures for software deployment, release, and operations, such as CI/CD pipelines.	18.4
Management	Development Process	Defines the development workflow, including guidelines for version control systems like Git.	65.1
	Project Management	Information related to the planning, organization, and management of the project.	5.4
Quality	Maintenance	Guidelines for system maintenance, including strategies for improving readability, detecting and resolving bugs.	44.6
	Debugging	Explains error handling techniques and methods for identifying and resolving issues.	25.3
	Performance	Focuses on system performance, quality assurance, and potential optimizations.	14.5
	Security	Addresses security considerations, vulnerabilities, or best practices for the system.	14.8
	UI/UX	Contains guidelines or details concerning the user interface (UI) and user experience (UX).	8.7

Below, we describe each category in detail and introduce a representative example.

System Overview. These instructions provide a general project or system overview and describe key features. We observed these instructions in 59.0% of files. Listing 4 shows an example.¹⁴ This example describes the project’s identity, mechanism, and functionality.

Listing 4. Example Instruction for Project Overview

```
1 ## Project Overview
2 Brightroom is a composable image editor library for iOS,
3 powered by Metal for high-performance image processing.
4 It provides both low-level image editing capabilities
5 and high-level UI components.
```

AI Integration. This instruction type contains notes for integrating or interacting with agentic coding tools. We observed these instructions in 24.1% of files. Listing 5 shows an example.¹⁵ This example provides a specific role and expertise for the agent, defining it as an expert in React, Next.js, and content management for technology community websites and blogs. It specializes in helping with the Craft Code Club website and blog platform, offering guidance on development, content management, SEO optimization, and performance tuning.

¹⁴<https://github.com/FluidGroup/Brightroom/blob/main/CLAUDE.md>
¹⁵<https://github.com/craft-code-club/blog-c3/blob/main/.github/copilot-instructions.md>

Listing 5. Example Instruction for AI Integration

```
1 # React & Next.js Community Technology Blog Expert Profile
2 You are an expert in React, Next.js, and content management for technology community
  websites and blogs. You specialize in helping with the Craft Code Club website and blog
  platform, providing guidance on development, content management, SEO optimization, and
  performance tuning for this software engineering community.
```

Documentation. This instruction type lists supplementary documents, links, or references for additional context. We observed these instructions in 26.8% of files. Listing 6 shows an example.¹⁶ This example provides specific documentation requirements: adding inline comments to explain non-obvious code, including JSDoc comments for public APIs, documenting complex types with examples, and providing usage examples in hook file comments.

Additionally, Listing 7 demonstrates how to document updates to the agent context files itself, instructing AI agents to update the file whenever they learn new project information that future tasks might need, noting that keeping guidelines current helps everyone work more effectively.¹⁷

Listing 6. Example Instruction for Documentation

```
1 ### Documentation Requirements
2 - Complex Logic: Add inline comments explaining non-obvious code
3 - Public APIs: Include JSDoc comments for exported functions
4 - Type Definitions: Document complex types with examples
5 - Hook Usage: Provide usage examples in hook file comments
```

Listing 7. Example Instruction for Documenting how to update agent context files themselves

```
1 Updating this document
2 AI agents should update this file whenever they learn something new about this project
  that future tasks might need to take into account. Keeping the guidelines current helps
  everyone work more effectively.
```

Architecture. This instruction type describes the high-level structure, design principles, or key components of the system’s architecture. We observed these instructions in 68.1% of files. Listing 8 shows an example.¹⁸ This example provides architectural structure, defining the project as a pnpm workspace with several packages under src/. It details the roles of packages such as core (shared business logic in TypeScript), background (initializes the extension store and wires services). It mandates that business logic reside in src/core and be implemented as use cases orchestrated through epics, and that UI components come from src/design-system/src.

Listing 8. Example Instruction for Architecture

```
1 ## Architecture
2 - The project is a pnpm workspace with several packages under `src/`:
3   - `core` - shared business logic implemented in TypeScript. Source files live in `src/
  core/src` and compile to `src/core/lib`.
4   - `background` - initializes the extension store and wires services such as `IndexedDBFS
  `, `FetchLoader` and `M3u8Parser`.
5   (continue...)
6 - Business logic should reside in `src/core`. Implement new features as
7   `use-cases` under `src/core/src/use-cases` and orchestrate them through epics
8   in `src/core/src/controllers`. Background scripts should only coordinate these
```

¹⁶<https://github.com/skip-mev/skip-go/blob/main/AGENTS.md>
¹⁷<https://github.com/zoonek/zoonek/blob/main/AGENTS.md>
¹⁸<https://github.com/puemos/hls-downloader/blob/master/AGENTS.md>

```
9 functions.
10 - UI components should come from `src/design-system/src` to keep styling
11 consistent across the extension.
```

Implementation Details. This instruction type provides specific details for implementing code or system components, including coding style guidelines. We observed these instructions in 70.8% of files. Listing 9 shows an example.¹⁹ This example provides code style guidelines, such as keeping code simple, explicit, typed, test-driven, and ready for automation. It also specifies constraints like using only ASCII characters in source files, the format for docstrings, preferring explicit constructs (e.g., no wildcard imports), referencing modules by alias, and using decorators like `@beartype.beartype`. Furthermore, it details naming conventions for classes (CamelCase), functions/variables (snake_case), constants (UPPER_SNAKE), and file descriptors/paths.

Listing 9. Example Instruction for Implementation Details

```
1 # Code Style
2 - Keep code simple, explicit, typed, test-driven, and ready for automation.
3 - Source files are UTF-8 but must contain only ASCII characters. Do not use smart quotes,
  ellipses, em-dashes, emoji, or other non-ASCII glyphs.
4 - Docstrings are a single unwrapped paragraph. Rely on your editor's soft-wrap.
5 - Prefer explicit over implicit constructs. No wildcard imports.
```

Build and Run. This instruction type outlines the process for compiling source code and running the application, often including key commands. We observed these instructions in 63.0% of files. Listing 10 shows an example.²⁰ This example provides specific npm build commands, such as `npm run dev` (for WXT dev server with live reload), `npm run dev:firefox` (for Firefox MV2 dev session), `npm run build` (for production build), and `npm run build:firefox` (for building and packaging for Firefox).

Listing 10. Example Instruction for Build and Run

```
1 ## Build Commands
2 - `npm run dev` - WXT dev server with live reload for Chromium targets (runs manifest
  update first)
3 - `npm run dev:firefox` - Firefox MV2 dev session (`wxt --browser firefox --mv2`; opens a
  temporary private profile)
4 - `npm run build` - Production build (validates locale files, copies ONNX files, updates
  manifest)
5 - `npm run build:firefox` - Build and package for Firefox
```

Testing. This instruction type details the procedures and commands for executing automated tests. This was the most prevalent category. We observed these instructions in 75.9% of files. Listing 11 shows an example.²¹ This example provides guidelines for writing tests using DOM and user interactions rather than internal implementation details, commands for running different test suites with their execution times, and warnings about environment-specific failures and testing constraints.

Listing 11. Example Instruction for Testing

¹⁹<https://github.com/samuelstevens/biobench/blob/main/AGENTS.md>

²⁰<https://github.com/Pedal-Intelligence/saypi-userscript/blob/main/CLAUDE.md>

²¹<https://github.com/SwitchbackTech/compass/blob/main/AGENTS.md>

```

1 ### Testing
2 ### Writing Tests in `@compass/web`
3 - Write tests the way a user would use the application by using the DOM and user
  interactions with `@testing-library/user-event` rather than internal implementation
  details of React components.
4 - Do NOT use `data-` attributes or CSS selectors to locate elements. Use semantic locators
  and roles instead.
5 ##### Running Tests
6 - **Core tests**: `yarn test:core` - Takes ~2 seconds. NEVER CANCEL. Always tests pass.
7 - **Web tests**: `yarn test:web` - Takes ~15 seconds. NEVER CANCEL. All tests pass

```

Configuration and Environments. This instruction type provides instructions for configuring the system and setting up the development or production environment. We observed these instructions in 38.9% of files. Listing 12 shows an example.²² This example provides commands for environment setup, including setting up git configuration (“cp example.gitconfig .git/config”), creating and activating a virtual environment (“uv venv” and “source .venv/bin/activate”), and installing development dependencies (“uv pip install -e “.[dev]”).

Listing 12. Example Instruction for Configuration and Environments

```

1 ## Development Commands
2 ### Environment Setup
3 ```bash
4 # Set up git configuration (recommended for development)
5 cp example.gitconfig .git/config
6 git config user.name "Your Name"
7 git config user.email "your.email@example.com"
8 # Create and activate virtual environment
9 uv venv
10 source .venv/bin/activate # Linux/macOS
11 .\.venv\Scripts\Activate.ps1 # Windows
12 # Install dependencies with development tools
13 uv pip install -e ".[dev]"

```

DevOps (Deployment and Operation). This instruction type covers procedures for software deployment, release, and operations, such as CI/CD pipelines. We observed these instructions in 18.4% of files. Listing 13 shows an example.²³ This example provides information on CI Parity, detailing the CI pipeline defined in .github/workflows/ci.yml and listing various checks included, such as cargo check, cargo test, cargo clippy -D warnings, and Typos check.

Listing 13. Example Instruction for DevOps

```

1 ## CI Parity
2 The CI pipeline is defined in `.github/workflows/ci.yml` and includes:
3 - `cargo check` for all targets
4 - `cargo test` on core crates
5 - `cargo fmt -- --check`
6 - `cargo clippy -D warnings`
7 - `cargo doc`
8 - Typos check (`crate-ci/typos`)

```

²²<https://github.com/basher83/ProxmoxMCP/blob/main/CLAUDE.md>

²³<https://github.com/EricLBuehler/mistral.rs/blob/master/AGENTS.md#ci-parity>

Development Process. This kind of instruction defines the development workflow, including guidelines for version control systems like Git. We observed these instructions in 65.1% of files. Listing 14 outlines an example of these kinds of instructions.²⁴ In this example, the project provides Commit & Pull Request Guidelines, specifying that commits must use the imperative mood and prefer Conventional Commits (e.g., feat:, fix:), requiring developers to run quality checks (just fmt, just mypy, just test) before pushing, and outlining requirements for PRs (summary, linked issues, screenshots, migration notes, and passing CI).

Listing 14. Example Instruction for Development Process

```
1 ## Commit & Pull Request Guidelines
2 - Commits: imperative mood; prefer Conventional Commits (e.g., `feat:`, `fix:`, `docs:`)
  with a clear scope.
3 - Before pushing: run `just fmt`, `just mypy`, and `just test`; ensure Django checks pass.
4 - PRs: include summary, linked issues, screenshots for UI changes, and migration notes
  when applicable. CI must be green.
```

Project Management. This kind of instruction relates to the planning, organization, and management of the project. We observed these instructions in 5.4% of files. Listing 15 outlines an example of these kinds of instructions.²⁵ In this example, the project provides guidance on the Backlog & Future Improvements stored in the backlog/ directory, detailing its purpose (storage for enhancement plans), the required format (Markdown files with structured plans and mandatory YAML front matter), and instructs users to reference backlog items to understand context and priorities.

Listing 15. Example Instruction for Project Management

```
1 ## Backlog & Future Improvements
2 The `backlog/` directory contains planned improvements and refactoring tasks:
3 - Purpose: Organized storage for enhancement plans, technical debt items, and future
  feature ideas
4 - Format: Markdown files with structured plans including problem statements, proposed
  solutions, and implementation strategies
5 - Front Matter: All backlog items MUST include YAML front matter with `status` (todo|in
  progress|done), `date_created`, and `date_modified` fields
6 (continue...)
7 - Usage: When suggesting improvements or picking up development work, reference backlog
  items to understand context and priorities
```

Maintenance. This kind of instruction provides guidelines for system maintenance, including strategies for improving readability, detecting and resolving bugs. We observed these instructions in 44.6% of files. Listing 16 outlines an example of these kinds of instructions.²⁶ In this example, the project provides Core Development Principles focused on maintaining stable public interfaces, specifically advising agents to always attempt to preserve function signatures, argument positions, and names for exported methods.

Listing 16. Example Instruction for Maintenance

```
1 ## Core Development Principles
2 ### 1. Maintain Stable Public Interfaces CRITICAL
```

²⁴<https://github.com/eduzen/website/blob/main/AGENTS.md>

²⁵<https://github.com/reinier/dotfiles/blob/main/CLAUDE.md>

²⁶<https://github.com/langchain-ai/langchain/blob/master/CLAUDE.md>

```

3 **Always attempt to preserve function signatures, argument positions, and names for
  exported/public methods.**
4 **Bad - Breaking Change:**
5 ```python
6 def get_user(id, verbose=False): # Changed from `user_id`
7     pass
8 ```
9 **Good - Stable Interface:**
10 ```python
11 def get_user(user_id: str, verbose: bool = False) -> User:
12     """Retrieve user by ID with optional verbose output."""
13     pass
14 ```

```

Performance. This kind of instruction focuses on system performance, quality assurance, and potential optimizations. We observed these instructions in 14.5% of files. Listing 17 outlines an example of these kinds of instructions.²⁷ In this example, the project provides performance guidelines for React 19, explaining when automatic compiler optimizations are sufficient versus when manual optimization is needed, and listing specific considerations for maintaining optimal performance.

Listing 17. Example Instruction for Performance

```

1 ## Performance Guidelines
2 ### React 19 Optimizations
3 - **Automatic optimizations**: React 19 compiler handles most memo/callback
4   optimizations
5 - **When to still optimize manually**:
6   - Heavy computational functions inside components
7   - Complex object/array transformations
8     (continue...)
9 ### Performance Considerations
10 - Avoid inline object/array creation in render
11 - Avoid function components defined inside a component or other unstable
12   functions
13 - Use stable references for callbacks passed as props

```

Security. This kind of instruction addresses security considerations, vulnerabilities, or best practices for the system. This was one of the least prevalent categories, observed in only 14.8% of files. Listing 18 outlines an example of these kinds of instructions.²⁸ In this example, the project provides details on the Security Architecture and Permission System. It highlights core principles like Workspace Isolation and Default Protection for preventing unauthorized access, and defines different Service Visibility Levels with their access permissions.

Listing 18. Example Instruction for Security

```

1 ## Security Architecture and Permission System
2 ### Core Security Principles
3 Hypa implements a multi-layered security model with workspace isolation as the foundation.
  All services have default protection through workspace visibility settings, with
  additional fine-grained permission controls available for sensitive operations.
4 ### Default Protection: Workspace Isolation

```

²⁷<https://github.com/hedoluna/fft/blob/main/CLAUDE.md>

²⁸<https://github.com/amun-ai/hypa/blob/main/CLAUDE.md?plain=1>

```
5 By default, all services are `protected`, meaning they are only accessible to clients
  within the same workspace. This provides automatic protection against unauthorized cross-
  workspace access without requiring explicit permission checks in every method.
6 (continue...)
7 ### Service Visibility Levels
8 1. `protected` (DEFAULT) - Only accessible by clients in the same workspace
9 2. `public` - Accessible by all authenticated users across workspaces
10 3. `unlisted` - Same as public, accessible for all users, but not discoverable
```

UI/UX. This kind of instruction contains guidelines or details concerning the user interface (UI) and user experience (UX). This was the second least prevalent category overall, observed in only 8.7% of the files. Listing 19 outlines an example of these kinds of instructions.²⁹ In this example, the project provides User Experience Standards, mandating an “invisible, Netflix-like user experience” where data loading and processing occur automatically in the background. The key principles include automatic data handling, subtle notifications, progressive loading, and eliminating the need for manual user intervention.

Listing 19. Example Instruction for UI/UX

```
1 ## User Experience Standards
2 This project follows an invisible, Netflix-like user experience where data loading and
  processing happens automatically in the background. Key principles:
3 1. Database-first: Always query cached data before API calls
4 2. Auto-detection: Automatically detect and fix data quality issues
5 3. Subtle notifications: Keep users informed without interrupting workflow
6 4. Progressive enhancement: Core functionality works immediately, enhanced features load
  in background
7 5. No manual intervention: Users never need to click "Load Data" or understand technical
  details
```

Debugging. This kind of instruction is one of the new categories added during the analysis because many cases could not be categorized into existing labels. We observed these instructions in 25.3% of files. Listing 20 outlines an example of these kinds of instructions.³⁰ In this example, the project provides Debugging Tips, specifically commands like using `DEBUG=1` for verbose output, checking `error.log` for errors, using `RUST_BACKTRACE=1` for stack traces, and profiling with `cargo flamegraph` for performance analysis.

Listing 20. Example Instruction for Debugging

```
1 ### Debugging Tips
2 - Use `DEBUG=1` for verbose output
3 - Check `error.log` for detailed errors
4 - Use `RUST_BACKTRACE=1` for stack traces
5 - Profile with `cargo flamegraph` for performance
```

While instruction types are shared, agents show largely consistent distributions with minor platform-specific variations. All three coding agents receive instructions across all categories, indicating that developers require similar types of guidance regardless of the agent they use. As shown in Table 4, the top categories (e.g., Testing, Implementation Details, Architecture, and Build and Run) remain prominent for all agents, and the bottom categories (e.g., Debugging,

²⁹<https://github.com/bdougie/contributor.info/blob/main/CLAUDE.md>

³⁰<https://github.com/probelabs/probe/blob/main/CLAUDE.md>

Table 4. Instruction Category Distribution by Agent Type (%)

Category	Codex	Claude	Copilot
Architecture	10.55%	11.20%	10.87%
Build and Run	9.23%	12.04%	8.44%
Implementation Details	12.24%	10.13%	12.16%
Testing	13.94%	12.16%	10.87%
Development Process	12.62%	9.30%	10.16%
System Overview	7.72%	9.42%	10.87%
Maintenance	7.53%	7.99%	5.87%
Configuration & Env.	6.40%	6.91%	5.29%
Debugging	3.01%	3.93%	5.01%
AI Integration	3.39%	3.46%	4.72%

AI Integration) are consistently less frequent. The differences in percentages across agents are generally small, typically within 1% to 3%.

While there are minor variations, our statistical tests confirm that Claude Code places a higher emphasis on Build and Run instructions compared to others (medium effect size, $d = -0.34$ vs OpenAI Codex and $d = -0.36$ vs GitHub Copilot), suggesting that Claude Code is frequently used in environments where the agent is expected to actively execute and verify the build. Similarly, OpenAI Codex receives a slightly higher proportion of Testing (13.94%) and Development Process (12.62%) instructions compared to GitHub Copilot or Claude Code, and GitHub Copilot shows a higher density of System Overview (10.87%) and Debugging (5.01%) instructions compared to OpenAI Codex. However, these differences are not substantial enough to suggest fundamentally different usage patterns across agents. Rather, the overall similarity in distributions suggests that the nature of developer instructions is primarily driven by the tasks themselves, not by the specific agent being used.

Instructions directly govern the structural and stylistic properties of agent-generated code. Beyond mere categorization, we observe an alignment between the specific constraints defined in the agent context files and the implementation patterns in post-adoption commits. This suggests that agents actively parse and stick to the project-specific rules provided by developers. For instance, context file instructions often define strict naming conventions and architectural boundaries. Listing 21 shows an agent context file³¹ defining specific Python naming rules and a preference for keeping side effects isolated. Listing 22 demonstrates a corresponding AI-generated commit³² where the agent correctly implemented a batch processing loop using the requested UPPER_SNAKE_CASE for environment-derived constants and adhered to the specified logging and indentation styles.

Listing 21. Example Agent Context File

```
1 ## Coding Style & Naming
2 - Python >= 3.11; follow PEP 8; 4-space indentation.
3 - Names: modules/functions `snake_case`, classes `CapWords`, constants `UPPER_SNAKE_CASE`.
4 - Prefer type hints and small, focused functions. Keep side effects in orchestrators (`
  main.py`, `app/bootstrap.py`, `rpc.py`, `jobs.py`).
```

³¹<https://github.com/skhomuti/csm-sentinel/blob/main/AGENTS.md?plain=1>

³²<https://github.com/skhomuti/csm-sentinel/commit/1cd3b568a65b2f06fbddd2aa1e481c6308f07015>

Listing 22. Corresponding Code

```

1 for batch_start in range(start_block, current_block + 1, batch_size):
2     batch_end = min(batch_start + batch_size - 1, current_block)
3     for contract in [os.getenv("CSM_ADDRESS"), os.getenv("FEE_DISTRIBUTOR_ADDRESS"), os.
4         getenv("VEBO_ADDRESS"),]:
5         logger.debug("Fetching logs for %s blocks %s-%s", contract, batch_start, batch_end
6             ,)

```

To move beyond the illustrative examples, we conduct two case studies on a simple, machine-checkable rule: 4-space indentation (Listing 21). For each case, we measure indentation compliance in AI-authored commits before and after the rule was added to the context file. We focus on this instruction because the intent is unambiguous, compliance can be checked automatically on every added line, and projects that introduce the rule during development provide a natural before-and-after comparison. We identify candidate projects by searching context files for explicit references to 4-space indentation and select one project per agent with a clearly identifiable rule-introduction commit and sufficient AI-authored commits in both periods. In a repository³³ using Claude Code, AI-authored commits produced 208 violating lines out of 342 eligible added lines before the rule was introduced, a violation rate of 60.82%. Afterward, the rate dropped to 11.03% (939 of 8,516 lines), an absolute reduction of about 50% despite nearly 25 times more added lines. In a repository³⁴ using GitHub Copilot, the violation rate fell from 20.17% (6,673 of 33,080 lines) before the rule to 7.57% (5,306 of 70,060 lines) afterward, a reduction of about 13%. The violation rate decreased substantially in both repositories. Although two cases cannot establish causality, the consistent decline after introducing the rule supports our qualitative observations and suggests that explicit, machine-readable instructions can meaningfully influence code formatting.

Answer to RQ₃

We identified 16 instruction categories. Instructions for functional aspects (Build and Run, Implementation Details, Architecture, Testing) are more prevalent, while meta-level or non-functional categories such as Performance, Security, and UI/UX appear far less frequently.

4.4 RQ₄: To what extent can instructions in agent context files be classified automatically?

4.4.1 Motivation. The manual content analysis in Section 4.3 provides a detailed taxonomy of instructions in agent context files, but manual qualitative coding is inherently labor-intensive and not scalable. As the adoption of agentic coding tools accelerates, the volume of agent context files will grow, making manual inspection a bottleneck for monitoring ecosystem-wide trends. The motivation for automating this classification is to provide a research instrument that enables the study of agentic coding practices at scale. This automation benefits several key audiences. For researchers, it enables long-term studies to examine how instruction patterns evolve over time and facilitates the comparison of practices across different programming communities or project types without excessive manual effort. It also allows for the easy replication of empirical studies across larger, emerging datasets. For tool builders, understanding the distribution of instructions can inform the design of agent scaffolding tools, linters, or documentation templates that encourage

³³<https://github.com/rgerhards/rsyslog>

³⁴<https://github.com/NB-Core/lotgd>

Table 5. Performance of automatic classification using GPT-5 on agent context files. The highest value within each category is highlighted in **bold**.

Category	Label	Precision	Recall	F1-Score	# Support
General	System Overview	0.88	0.90	0.89	196
	AI Integration	0.33	0.86	0.48	80
	Documentation and References	0.43	0.87	0.57	89
Implementation	Architecture	0.89	0.97	0.93	226
	Implementation Details	0.85	0.93	0.89	235
Build	Build & Run	0.90	0.94	0.92	209
	Testing	0.91	0.96	0.94	252
	Configuration & Environment	0.64	0.91	0.75	129
	DevOps	0.64	0.84	0.72	61
Management	Development Process	0.92	0.76	0.83	216
	Project Management	0.40	0.44	0.42	18
Quality	Maintenance	0.65	0.50	0.56	148
	Debugging	0.70	0.73	0.71	84
	Performance	0.59	0.90	0.71	48
	Security	0.60	0.98	0.74	49
	UI/UX	0.48	0.69	0.56	29
Micro Average		0.73	0.86	0.79	2,069

developers to include comprehensive guidance. For practitioners, ecosystem-wide analysis can highlight common blind spots. The goal of this classification is not to feed results back into the LLMs for operational use, but to empower the software engineering community to detect emerging gaps, anti-patterns, and best practices in agent configuration. Recent work suggests that LLMs can effectively perform classification tasks typically reserved for human annotators [5]. Therefore, we investigate the possibility of automating the classification of agent context files to enable future large-scale monitoring.

4.4.2 Approach. We frame the problem as a multi-label classification task over the 16 categories from Section 4.3 (Table 3). We leverage GPT-5 to automatically classify the content of the 332 files from our manually labeled RQ3 dataset, with each request using *temperature* = 1. This dataset comprises a sample of Claude Code, OpenAI Codex, and GitHub Copilot files, providing a diverse representation of agent context instructions. By using this specific subset, we ensure that the model’s performance is measured against the high-quality consensus labels reached by our three human inspectors.

The model is formulated as a multi-label binary classifier. We construct a prompt containing (i) the full agent context file content and (ii) a concise definition of each category accompanied by representative examples. To make the model’s output directly consumable by our evaluation pipeline, we constrain it to produce predictions using a JSON structured-output schema. The prompt and implementation details are available in our replication package. These examples were selected based on the most frequently recurring patterns identified during the Phase 2 manual labeling in RQ3, rather than chosen at random. For instance, for the Architecture category, we included

snippets showing directory mappings as in Listing 8, as this was the most common structural pattern found in the ground truth.

To evaluate performance, we compare the model’s predictions against the human-established ground-truth labels from Section 4.3. We compute precision, recall, and F1-score per category, as well as the micro average across all 2,069 label assignments.

4.4.3 Findings. Automatic classification of agent context files is feasible and promising, with an overall micro-average F1-score of 0.79. As shown in Table 5, the model performs particularly well on concrete, functional instructions in the Implementation, Build, and Management categories. Labels such as System Overview, Architecture, and Testing achieve high F1-scores, indicating that instructions in these areas are distinct and consistently expressed, which allows the model to recognize their patterns reliably. Build and Run, Implementation Details, and Development Process also exhibit strong performance.

In contrast, the model struggles with categories that are more abstract or have semantically overlapping or sparsely represented examples. Within the Quality and General categories, labels such as Maintenance, Project Management, AI Integration, and Documentation and References obtain comparatively lower F1-scores. This suggests that, while the model is effective at detecting concrete commands and technical structures, it has more difficulty distinguishing nuanced or infrequent instructions related to project management or maintainability.

Answer to RQ₄

Automatic classification of agent context files is feasible, achieving a micro-average F1-score of 0.79. The model performs well on concrete functional categories such as Architecture and Testing, but shows weaker performance on more abstract or infrequent categories such as Maintenance.

5 IMPLICATIONS

In this section, we discuss the implications of our findings for researchers, developers, and coding agent builders.

5.1 Implications for researchers

Define and measure context debt as a new form of technical debt. Our analysis in Section 4.1 shows that agent context files are not merely configuration files but extensive documentation artifacts. In particular, Claude Code files exhibit a median Flesch reading ease (FRE) score of 42.39, which classifies them as difficult to read and comparable to high school-level material under the FRE interpretation bands. This low readability, combined with their considerable length, suggests that as projects evolve, these agent context files accumulate context debt: a state where instructions meant to clarify context for an AI agent become unmaintainable and opaque to human collaborators. This creates a paradox where a mechanism designed to align agents increases the cognitive load on developers. Future work should formalize context debt by identifying specific agent context file smells (e.g., ambiguous directives, conflicting role definitions) and by developing metrics beyond standard readability scores to assess the maintainability of agent instructions.

Investigate the granular impact of structural elements on agent performance. Our findings in Section 4.1 reveal that developers rely on a consistent, shallow hierarchy to organize agent context. It remains an open research question whether removing these H1 and H2 headers, or altering their hierarchy, would significantly degrade an LLM’s comprehension or its ability to execute tasks accurately. Recent studies report that agent context files fundamentally alter

agent efficiency and repository exploration behavior [23, 38]. Future work should move beyond binary presence or absence evaluations and conduct controlled ablation studies on the structural markers themselves. Quantifying how formatting, hierarchical depth, and header removal impact downstream programming tasks will clarify if current developer practices optimize or inadvertently hinder agentic workflows.

Model the co-evolution of agent context files and code to automate maintenance. Contrary to the write-once nature of traditional READMEs [22], our findings in Section 4.2 show that agent context files are actively maintained, with 67.0% of Claude Code files undergoing multiple modifications. These updates occur at day-scale intervals (median 23.8 hours for Claude Code, 68.0 hours for GitHub Copilot) and involve incremental additions rather than deletions. This active, ongoing maintenance pattern suggests that developers revisit agent context files as they encounter new project requirements over the course of regular development, which is consistent with a co-evolutionary relationship between agent context files and the underlying codebase. However, our current data do not directly trace which code changes trigger which agent context file updates. Future work should analyze paired code-and-agent context file commits to establish a more direct co-evolutionary link and quantify the degree of coupling. Building on such evidence, researchers could develop CI-integrated tools that automatically detect divergences. For example, a linter for agent context files could verify that the commands listed in the Build and Run section (63.0% prevalence) match the actual scripts in package.json or Makefile, which would reduce the manual effort required for synchronization.

Investigate the blind spots of agentic coding and build related benchmarks. Our content analysis (Section 4.3) shows that functional categories like Build and Run and Implementation Details dominate, while non-functional requirements (NFRs) such as Security (14.8%) and Performance (14.5%) are notably rare.

However, we note that the low prevalence of NFR instructions does not necessarily mean that developers neglect these concerns altogether. Developers may address security and performance through other means, such as manual code reviews, external tooling (e.g., static analysis, CI/CD security scanners), or by gradually adding NFR-related instructions in later commits. Our current analysis captures the presence of instruction categories at a single snapshot but does not track how the semantic content of individual commits evolves over time. Identifying when developers incorporated NFR instructions would require applying the automatic classification approach from RQ4 to each commit diff, which demands considerable effort and computational resources, and is left as future work. Nonetheless, the current gap in agent-facing NFR guidance remains noteworthy. If benchmarks (e.g., SWE-bench [30]) do not penalize agents for generating insecure or inefficient code, this blind spot will persist. The research community should design new benchmarks or evaluation frameworks that explicitly test an agent's adherence to NFRs defined in agent context files. Future work should also examine the post-generation activities of developers to understand whether and how they compensate for the absence of NFR instructions through testing, code review, or complementary tooling.

5.2 Implications for developers

Adopt a configuration-as-code mindset for agent context files. The active maintenance patterns observed in Section 4.2 indicate that agent context files behave more like dynamic configurations than static text. Developers should therefore treat files like CLAUDE.md or AGENTS.md with the same rigor applied to Dockerfiles or CI/CD workflows. We recommend integrating agent context file updates into the standard code review process. When a pull request modifies the build system or refactors a core module, the review checklist should explicitly ask whether the change

requires an update to the agent context files. Treating these files as living code artifacts prevents context drift and keeps the agent’s operational knowledge aligned with the current state of the project.

Explicitly include non-functional requirements (NFRs) to prevent quality degradation.

The scarcity of instructions related to Security and Performance (Section 4.3) serves as a warning.

This challenge may be compounded by the fact that writing effective LLM instructions is inherently difficult, even for motivated users [82], which could lead developers to focus on familiar functional topics while omitting less tangible quality attributes.

Unlike human senior developers who may implicitly understand secure coding practices, AI agents rely heavily on explicit context. If security guidelines are absent from the agent context file, agents may produce functional yet vulnerable code (e.g., SQL injection risks). Developers should proactively structure their agent context files to include mandatory sections for NFRs. Directives such as “All database interactions must use parameterized queries” or “Do not commit secrets” should be codified within the agent context file to act as continuous guardrails during generation.

Implement versioning and governance for agent context files’ evolution. Given that agent context files grow through frequent, small additions (Section 4.2), they risk becoming unstructured append-only logs. To address this, developers should apply semantic versioning to their agent context files and maintain a changelog. Because agents may misinterpret instructions, changes to high-impact sections like Architecture or Development Process should require approval from a designated context owner via CODEOWNERS,³⁵ ensuring that the agent’s behavioral guidelines remain consistent and authoritative.

5.3 Implications for coding agent builders

Leverage structural consistency for scaffolded authoring tools. Our structural analysis in Section 4.1 shows that agent context files follow a consistent, shallow hierarchy based on H1 and H2 headers. Tool builders can use this pattern to lower the barrier to entry for developers. Instead of presenting an empty text editor, IDEs and agent interfaces should provide templated scaffolds pre-populated with the common categories identified in Section 4.3 (e.g., Build and Run, Implementation Details, Testing). These templates should also include placeholders for frequently neglected NFR categories (Security, Performance), which would encourage more comprehensive documentation.

Optimize context retrieval using semantic categorization. The categorization of instructions (Table 3 in Section 4.3) indicates that agent context files contain distinct clusters of information. Retrieval-augmented generation (RAG) systems for coding agents should move beyond naive text chunking and use this semantic structure. When an agent is tasked with fixing a bug, for example, retrieval should prioritize sections labeled Debugging (25.3%) and Testing, while deprioritizing unrelated sections like DevOps. By parsing the hierarchy of an agent context file, agent builders can improve the signal-to-noise ratio of the context supplied to the LLM, which should lead to more accurate and relevant code generation.

6 RELATED WORK

In this section, we discuss related work on AI agents in software engineering, software documentation, and context engineering.

³⁵<https://docs.github.com/en/repositories/managing-your-repositorys-settings-and-features/customizing-your-repository/about-code-owners>

6.1 AI agents in software engineering

AI in software development is shifting from single-turn code completion to agentic software engineering [28]. In this paradigm, agents understand high-level goals, produce plans, and execute multi-step tasks with minimal human intervention [11, 12, 61, 74]. These agents combine an LLM-based reasoning core with memory, planning modules, and tool use, such as interacting with IDEs, running tests, and invoking external services [18, 42]. This reframes LLMs from autocomplete engines into components within larger systems that must be configured, orchestrated, and maintained [31, 79]. Architectures are converging on perception, memory, and action patterns, where agents ingest context, ground decisions in persistent memory, and act through tool APIs [74].

Research has expanded from simple code generation to agents that can interactively clarify ambiguous requirements [16, 45, 77], refactor code [8, 29], and assess code quality [81]. Collaboration mechanisms further boost effectiveness. For example, multi-agent pair programming splits strategic planning and tactical coding across coordinated roles, such as Navigator and Driver agents [83], and scalable human-in-the-loop frameworks that keep developers in control of plans and reviews while letting agents draft and iterate code [48, 70]. Existing studies primarily focus on the agent's internal architecture and collaborative behavior, that is, how the agents reason and coordinate, rather than how they are configured from the outside.

Automatic program repair (APR) has been an early beneficiary of this paradigm. Base LLMs already rival or surpass traditional APR pipelines on several benchmarks [17, 24, 78], while agenzitized repair further improves reliability by iterating between code changes, test execution, and analysis [11]. Related work integrates specification extraction and formal feedback into the loop so that agents can better infer intent and refine patches [16, 45, 59]. Beyond repair and generation, agents have been proposed for autonomous testing [18] and code quality support, such as detecting smells and providing maintainability guidance [81].

With greater integration comes new failure modes. Empirical studies report integration defects and security risks across agent, vector store, connector, and execution layers [62], and recent work has examined how developers perceive and intervene on LLM-generated code in real projects [55]. For example, the Model Context Protocol (MCP) helps agents integrate with tools but introduces security risks [27]. At the same time, evaluating agents remains costly and variable. Comprehensive test-based assessments are expensive, while LLM-as-a-judge evaluations can be inconsistent. Real-world utilization depends on collaboration patterns and process fit [48, 69, 70]. These observations highlight the need for reliable artifacts and practices that communicate project-specific constraints to agents and keep behavior aligned over time.

Our work targets this missing layer: persistent configuration and context artifacts that steer agent behavior in real-world projects. Prior work on agentic software engineering largely concentrates on agents' internal architectures, tool-use loops, and coordination. When external steering is discussed, it is often framed as prompt engineering, i.e., task-level instructions crafted for a single interaction [63, 84]. In contrast, we examine project-level documents that configure agents from the outside and are automatically loaded at the start of each session, namely agent context files such as `CLAUDE.md`, `AGENTS.md`, and `copilot-instructions.md`. Rather than being discarded after one task, these context files accumulate roles, constraints, build and test routines, and coding conventions over time, and must be maintained alongside the codebase. While augmented language model designs advocate the use of external tools, memory, and retrieval to ground decisions [42], the community lacks a systematic understanding of how open-source projects encode such persistent guidance for agents at scale. To our knowledge, this is the first empirical analysis of these files, quantifying their structure, maintenance, and instruction taxonomies to complement existing work with evidence about this configuration layer.

6.2 Software documentation

Empirical research on software documentation spans a wide range of artifacts, such as READMEs, API documentation, issue discussions, and release notes. Prior studies found that README files are typically concise and function-focused, often expanding over time [22]. Over 90% of READMEs mention basic information such as project name, description, and usage instructions [51]. Large-scale repository mining has uncovered recurring documentation issues and antipatterns in APIs [2], fragmented feature documentation in popular GitHub projects [52], and patterns in how refactoring needs and improvement activities are described and tracked through issues [7]. Surveys and interviews further identify what practitioners value and which issues they consider most critical [3, 9].

Researchers have reported documentation issues found in software projects [4], but evaluating documentation quality remains challenging. Automatic metrics adapted from machine translation and summarization, such as BLEU, ROUGE, and METEOR, are widely used to assess generated comments and summaries [25, 40, 53]. However, several studies show that improvements in these metrics often fail to align with human-perceived usefulness, with only moderate correlations and even ranking inversions relative to practitioner judgments [32, 58, 65].

Beyond static quality, research also examines how documentation evolves. Studies on code-comment co-evolution show frequent inconsistencies [75], and techniques have been developed to detect outdated or contradictory statements in API documentation and narratives [71, 86]. Dynamic and hybrid approaches leverage runtime values or execution traces to generate or validate documentation, showing benefits for specific method categories [67, 68]. Recent datasets explicitly target code-documentation alignment during maintenance, underscoring that synchronization remains a persistent challenge [47].

Documentation concerns also surface in code review. Large-scale analyses show that reviewers routinely discuss, request, and amend documentation, making review a key gatekeeper for documentation quality [54]. Automated assistance increasingly participates in this process: review bots influence project throughput and communication patterns [76], and specialized frameworks detect and repair defects in API directives, including parameters and exceptions, with high precision [86].

Prior work has primarily investigated developer-facing documentation, such as READMEs, API references, comments, and review discussions, focusing on its quality, evolution, and automation. In contrast, we study agent-facing, persistent configuration artifacts (agent context files) that encode project context, roles, and operating rules for coding agents. Because their primary audience is an AI agent, these files often use imperative phrasing and explicit constraints, and they tend to be tightly coupled to day-to-day development workflows, including build, test, and contribution processes, and therefore to code evolution, which we quantify in RQ2. Moreover, our taxonomy of 16 instruction categories (RQ3) surfaces content such as AI integration guidance and explicit role definitions that has little counterpart in traditional documentation studies. To our knowledge, these agent context files have not been examined at scale in the documentation literature cited above.

6.3 Context engineering

A growing body of work investigates context engineering, which extends beyond traditional prompt engineering to optimize how developers provide information to AI coding agents. While prompt engineering typically optimizes transient, task-level interactions, context engineering concerns persistent artifacts such as configuration files, documentation, and contextual instructions that continuously shape agent behavior across sessions and projects. Researchers are empirically examining how different instruction strategies affect agent performance on diverse software engineering

tasks, including code generation [63], automated program repair [78, 80], and repository-level code completion [64].

This research can be broadly divided into two complementary areas: (a) the study of immediate, task-specific prompts, and (b) the study of persistent, project-level context. Much of the existing literature focuses on optimizing the immediate instructions that developers provide for specific, in-the-moment tasks. These include reasoning-oriented techniques like chain-of-thought (CoT) for multi-step tasks [49], interactive methods involving reflection and dialogue [44, 84], and analyses of recurring patterns in developers’ conversational prompts [33]. Other studies examine knowledge-enriched prompts that embed domain-specific data, such as UML model constraints [1] or exception handling patterns [56], to produce more robust and reliable outputs.

Complementing this, other work highlights that a primary failure mode for AI assistants is a fundamental “lack of contextual awareness” [6, 73], a problem that cannot be solved by a single prompt alone. This challenge has driven the development of mechanisms that provide agents with persistent, repository-level context [26, 64]. These context-rich approaches integrate repository-level information such as imports, module hierarchies, or architectural metadata to improve code completion and adaptation [21, 64]. Studies consistently show that such fine-grained, knowledge-driven context yields substantial improvements in code quality, correctness, and reliability over simple zero-shot prompts [1, 46, 63]. As providing this context becomes a core development activity, managing these instructions as long-lived artifacts has emerged as a new software engineering challenge [34].

While prior studies have explored how different prompt- and retrieval-based context strategies affect downstream task quality [21, 33, 64], they typically treat context as an experimental input and evaluate its *effect* on model performance. Our work is complementary. Rather than measuring the effect of providing context, we study the long-lived artifacts that practitioners create to provide and manage that context in real repositories. This paper presents the first empirical study of agent context files (e.g., CLAUDE.md, AGENTS.md) as key artifacts of context engineering, analyzing their structure, content, and maintenance patterns to understand how developers curate the persistent instructions that guide agentic coding in practice.

7 THREATS TO VALIDITY

In this section, we discuss the threats to validity of our study about agent context files.

7.1 Internal Validity

Threats to internal validity concern factors internal to our study that could have influenced our results. Section 4.3 involves manual classification of content within the agent context files, which introduces potential human error and subjective bias. To address this risk, two inspectors examined the agent context files independently and carefully. This independent labeling achieved 80.3% agreement, and a third inspector resolved any conflicting labels.

7.2 Construct Validity

Threats to construct validity concern the extent to which the measures used truly reflect the concepts being studied. Our manual content classification approach categorized instructions into 16 distinct labels based on the presence of a topic. For instance, the category Implementation Details encompasses specific guidance for implementing code or system components, including coding style guidelines. The threat arises because this classification was purely binary; an agent context file was flagged with the Implementation Details label if it contained any mention of code style, regardless of whether that content was minimal (one line of instruction) or substantial (many lines detailing complex conventions). Consequently, the frequency reported for a category represents

only the prevalence of the topic, not the depth, complexity, or qualitative richness that developers invested in that specific instruction set.

To confirm the reliability of our programmatic approach for assessing readability in technical artifacts, we cross-checked our findings using an alternative framework, `py-readability-metrics`³⁶. Across the studied agent context files, the Mann-Whitney U test confirms no significant difference between the tools across the studied files (median difference of 0.279), demonstrating the robustness of Textstat. Manual inspection of the minor variations showed that they stem from expected differences in parsing non-prose elements. Textstat treats newlines and bullet points as sentence boundaries, appropriately capturing the structure of list-heavy technical files, whereas `py-readability-metrics` strictly requires terminal punctuation. Additionally, slight fluctuations arise from differences in syllable estimation techniques for technical strings such as file paths. Overall, these parsing differences are marginal and do not compromise the reliability of Textstat.

Another construct validity concern applies to the use of FRE as a readability proxy for agent context files. FRE was designed and validated for general English prose, and its two underlying assumptions, that longer sentences and higher syllable counts per word indicate greater comprehension difficulty, do not map cleanly onto technical, procedural, and code-adjacent text. As our manual inspection revealed, FRE scores are strongly influenced by content form rather than true cognitive difficulty for the intended audience. Files composed of short imperative directives and inline command tokens receive high (easy) FRE scores because their surface-level lexical and syntactic properties mimic simple prose, yet they may be no easier to act upon for a developer unfamiliar with the codebase. Conversely, files providing rich architectural context with stacked technical noun phrases, framework names, and acronym expansions receive low (hard) FRE scores, even though experienced developers in the relevant domain may find such descriptions clear and actionable. Consequently, low FRE scores in our dataset may partly reflect technical vocabulary and document form rather than genuine comprehension difficulty. Researchers and developers should therefore treat FRE scores in this context as a measure of surface-level lexical and syntactic complexity rather than as a direct measure of usability or clarity for the developers who use these files.

A further concern relates to our dependence on Markdown header syntax as the basis for measuring the structural organization of agent context files. Markdown is flexible, and developers may use its structural elements inconsistently across projects. For example, some developers may use bold text or horizontal rules to define each section rather than explicit header tags, while others may use deeper header levels (e.g., H3, H4) to indicate what others would express as bullet points or paragraph emphasis. Our header-counting approach treats only lines beginning with one or more hash symbols followed by a space as headers, which means that visually structured but syntactically unmarked sections are not captured in our analysis. Consequently, the reported distributions of H1 through H5 headers reflect explicit Markdown usage patterns rather than the full range of structural conventions developers may employ. We chose this strict syntactic definition because it is standardized across all agent context files in our dataset, whereas inferring structure from visual or semantic cues would introduce subjective interpretation. Nonetheless, readers should interpret our structural findings as characterizing how developers use formal Markdown headers, rather than as a complete account of all organizational strategies present in agent context files. Future work could complement our analysis by examining alternative structural signals such as bolded section titles, horizontal rule separators, or list-based hierarchies.

³⁶<https://pypi.org/project/py-readability-metrics/>

7.3 External Validity

Threats to external validity concern the generalizability of our findings. This study examined 2,303 agent context files from 1,925 repositories that use one of three major agentic coding systems (*i.e.*, Claude Code, OpenAI Codex, and GitHub Copilot). Although we expanded the number of studied files compared to our previous study [14], the dataset remains limited, which constrains the generalizability of our findings. Future work should extend this analysis to include a larger number of files from various agentic coding systems.

8 CONCLUSION

This study presented the first empirical analysis of the structure, maintenance, and content of agent context files, such as `CLAUDE.md`, `AGENTS.md`, and `copilot-instructions.md`, which play an important role in defining and operationalizing AI agent behavior in agentic coding. The primary motivation for this research was the significant lack of accessible documentation for creating these files, which has historically forced developers into inefficient trial-and-error approaches when configuring their agents. Our investigation systematically analyzed 2,303 context files collected from 1,925 open-source repositories to provide an empirical foundation for best practices.

Our findings regarding the characteristics of agent context files reveal that these files are generally long and difficult to read. Additionally, such files for Claude Code and GitHub Copilot are substantially longer than those for OpenAI Codex, suggesting developers provide a much larger volume of natural language instructions to these agents. Critically, the readability of these files is poor; the documents are complex, with many categorized as the “difficult” category comparable to high school-level material. Structurally, agent context files are consistently organized with a shallow hierarchy, anchored by a single H1 heading and using H2 and H3 subsections to define major topics, a pattern that likely aids developers in quickly parsing and maintaining the documents. Furthermore, unlike conventional software documentation often characterized as “write-once,” agent context files are actively maintained, behaving as evolving configuration artifacts. The majority of Claude Code files (67.0%) are revisited and refined across multiple commits, with maintenance occurring in short, rapid bursts. This evolution is driven by small, incremental content additions, while content deletions are negligible.

The analysis of instructional content demonstrated that developers primarily focus on action-oriented, functional guidance necessary for execution and maintenance. The most prevalent instructions address practical operations, including Testing (75.9%), Implementation Details (70.8%), and Architecture (68.1%). However, the study identified critical omissions: instructions addressing non-functional requirements (NFRs), such as Security (14.8%), Performance (14.5%), and UI/UX (8.7%), were notably infrequent. This suggests that agents are extensively guided on “how” to build the code functionally, but not necessarily on “how to build it well” concerning quality attributes.

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A APPENDIX

A.1 Mapping of Manifest Headers to Taxonomy Labels

Table 6 details the systematic mapping of original manual labels and representative headers found in the agent context files (GitHub Copilot, Claude Code, OpenAI Codex) to the final categories used in our qualitative analysis.

Table 6. Taxonomy Mapping for Agent Context File Content

Original Label	Example Header from Agent Context File	Final Merged Label
Decision	Go/No-Go Decision Points	Project Management
Administration	Task Management	Project Management
Algorithms and Data Structures	Search Flow	Implementation Details
API & Interface	Run API server	Build and Run
Architecture	Architecture	Architecture
Backend & API	API Key (Development)	Architecture
Best Practices	Dashboard Implementation Best Practices	Implementation Details
Bibliography	References	Documentation and References
Build & Test	Building and Running	Build and Run
Business Domain	The Business Logic Contract	System Overview
Business Logic	Business Needs	System Overview
CI/CD	CI/CD Pipeline	DevOps
AI Integration	Claude Code Configuration	AI Integration
Code Style	Code Style Guidelines	Implementation Details
Collaboration	Common Workflow Patterns	Development Process
Communication	Agent Communication Flow	Architecture
Component	Conversation Manager (conversationManager.js)	Architecture
Configuration	Environment Variables	Configuration and Environment
Context	Critical Context	System Overview
Data Management	Primary Data Sources	Architecture
Data Science and ML	Model breakdown display	Architecture
Data Structure	Data Structure Classes	Architecture
Database	Connect to database	Architecture
Dependencies and Libraries	Dependency Injection Setup	Architecture
Deployment and Operations	Production Deployment Details	DevOps
Development Commands	Build and Run Commands	Build and Run
Development Environment	Setup Docker Operations	Configuration and Environment
Development Timeline	Phase 2: Trust & Transparency	Project Management
Development Workflow	Bug Fix Workflow	Development Process
Documentation	Key Documentation References	Documentation and References
Error Handling & Debugging	Debugging & Troubleshooting	Debugging
Example	Code Examples	Documentation and References
Features	Camera System	System Overview
Implementation Details	Important Technical Details	Implementation Details
Infrastructure Configuration	Ansible	Configuration and Environment
Integration	Event System Integration	Architecture
Limitations	Known Limitations	System Overview
Logging	Logging Infrastructure	Debugging
Memory Management	Memory Usage	Performance
Network and Communication	Routing and Request Handling	Architecture
Notes	Technical Notes	Documentation and References
Operating System	VeridianOS Overview	System Overview
Overview	Application Overview	System Overview
Performance & QA	Performance and Monitoring	Performance
Project Management	Task Assessment Phase	Project Management
Project Operations	Sheet Operations	Implementation Details
Project Overview	Project Description	System Overview
Project Structure	Current Project Structure	Architecture
Project Workflow	Initiative Completion Flow	Development Process
Refactoring	Recent Improvements	Maintenance
Reminder	Auto-Push Reminder	Development Process
Security	Security Considerations	Security
Success Metrics	Success Criteria	Project Management
System Workflow	Execution Flow	Architecture
Task	Development TODOs	Project Management
Technology Stack	Framework Stack	Architecture
Tips	Development Tips	Development Process
Tools	Additional Tools	Development Process
UI/UX	User Experience	UI/UX
Version Control & Git	Git and Development Workflow	Development Process
Work Progress	Current Progress	Project Management